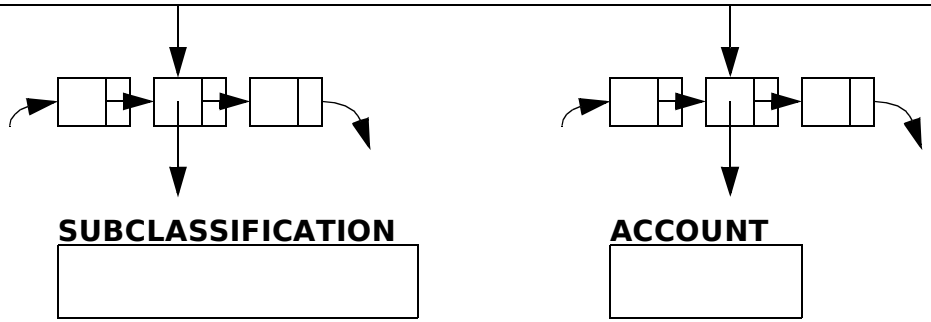
PredictiveBooks Data Structure

ELEMENT

```
/* Usage */
LIST *element_list(
        LIST *filter_element_name_list,
        char *fund_name,
        char *transaction_date_time_closing(),
        boolean fetch_subclassification_list,
        boolean fetch_account_list );

/* Attributes */
char *element_name;
boolean accumulate_debit;

/* Process */
int percent_of_asset;
int percent_of_revenue;
LIST *subclassification_list;
LIST *account_list;

void element_list_account_action_string_set(
        LIST *element_list,
        char *application_name,
        char *session,
        char *login_name,
        char *role_name,
        char *begin_date_string,
        char *as_of_date );

void element_list_total_set(
        LIST *element_list );

void element_list_percent_of_asset_set(
        LIST *element_list );

void element_list_percent_of_revenue_set(
        LIST *element_list );

void element_denominator_percent_of_asset_set(
        LIST *element_list,
        double asset_total );

void element_denominator_percent_of_revenue_set(
        LIST *element_list,
        double revenue_total );

void element_list_delta_prior_set(
        LIST *prior_year_element_list,
        LIST *preclose_element_list );

void element_prior_year_element_list_delta_prior_set(
        LIST *prior_year_element_list,
        ELEMENT *preclose_element );

int element_delta_prior(
        double prior_total,
        double total );

void element_list_balance_total(
        LIST *element_list );

double element_balance_total(
        LIST *subclassification_list,
        LIST *account_list );

double element_debit_total(
        LIST *element_list );

double element_credit_total(
        LIST *element_list );

ACCOUNT *element_list_account_seek(
        char *account_name,
        LIST *element_list );
```



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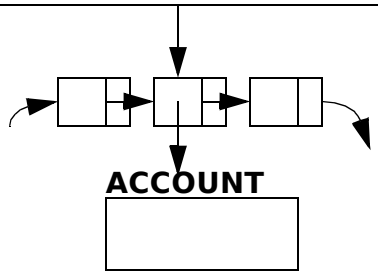
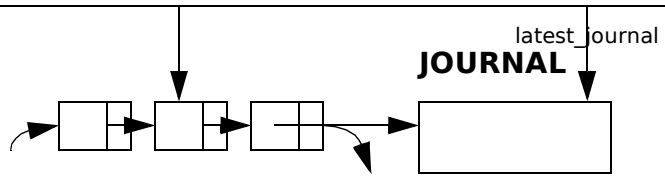
SUBCLASSIFICATION

```
/* Usage */
SUBCLASSIFICATION *subclassification_total_fetch(
        double *subclassification_total,
        char *subclassification_name,
        char *fund_name,
        char *transaction_date_time_closing() );

/* Attributes */
char *subclassification_name;

/* Process */
double subclassification_total;
int percent_of_asset;
int percent_of_revenue;

LIST *account_list =
        subclassification_total_account_list(
            &subclassification_total,
            subclassification_name,
            fund_name,
            transaction_date_time_closing() );

void subclassification_list_percent_of_asset_set(
        LIST *subclassification_list,
        double asset_total );

void subclassification_list_percent_of_revenue_set(
        LIST *subclassification_list,
        double revenue_total );

void subclassification_list_delta_prior_set(
        LIST *prior_subclassification_list,
        LIST *subclassification_list );

void subclassification_delta_prior_set(
        LIST *prior_subclassification_list,
        SUBCLASSIFICATION *subclassification );

int delta_prior =
        element_delta_prior(
            prior_subclassification->subclassification_total,
            subclassification->subclassification_total );

double subclassification_total =
        subclassification_balance_total(
            subclassification_list );

double subclassification_debit_total(
        LIST *subclassification_list );

double subclassification_credit_total(
        LIST *subclassification_list );

LIST *subclassification_account_list(
        LIST *subclassification_list );
```



ACCOUNT

```
/* Attributes */
char *account_name;
char *fund_name;
char *subclassification_name;
char *hard_coded_account_key;
boolean accumulate_debit;
double balance;
double payment_amount;

/* Process */
LIST *journal_list;
JOURNAL *latest_journal;
int percent_of_asset;
int percent_of_revenue;

char *account_action_string(
        char *application_name,
        char *session,
        char *login_name,
        char *role_name,
        char *begin_date_string,
        char *as_of_date,
        char *account_name );

void account_list_percent_of_asset_set(
        LIST *account_list,
        double asset_total );

void account_list_percent_of_revenue_set(
        LIST *account_list,
        double revenue_total );

void account_list_delta_prior_set(
        LIST *prior_account_list,
        LIST *account_list );

void account_delta_prior_set(
        LIST *prior_account_list,
        ACCOUNT *account );

int delta_prior =
        element_delta_prior(
                prior_account->
                        latest_journal->
                        balance,
                account->
                        latest_journal->
                        balance );

double account_total =
        account_balance_total(
                LIST *account_list );

double account_debit_total(
        LIST *account_list );

double account_credit_total(
        LIST *account_list );

char *account_element_name(
        subclassification_name );
```

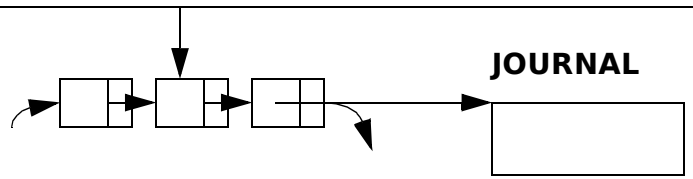


JOURNAL

```
char *full_name;
char *street_address;
char *transaction_date_time;
char *account_name;
double previous_balance;
double previous_balance_database;
double debit_amount;
double credit_amount;
double balance;
int transaction_count;
boolean match_sum_taken;
char *transaction_memo;

JOURNAL *journal_new(
        full_name,
        street_address,
        transaction_date_time,
        account_name );

char *journal_select(
        void );

JOURNAL *journal_parse(
        input );

LIST *journal_list(
        full_name,
        street_address,
        transaction_date_time );

LIST *journal_list_fetch(
        where_clause );

JOURNAL *journal_prior(
        transaction_date_time,
        account_name );
FILE *journal_insert_open(
        void );

void journal_insert(
        insert_pipe,
        full_name,
        street_address,
        transaction_date_time,
        account_name,
        amount,
        is_debit );

/* Also does journal_propagate() for each account */
void journal_delete(
        full_name,
        street_address,
        transaction_date_time );

void journal_propagate(
        transaction_date_time,
        account_name );

LIST *journal_list_minimum(
        minimum_transaction_date_time,
        account_name );

LIST *journal_list_account(
        account_name );

void journal_list_set_balances(
        journal_list_prior(),
        account_accumulate_debit(
            account_name ) );

void journal_list_propagate_update(
        journal_list_prior() );

JOURNAL *journal_account_fetch(
        account_name,
        transaction_date_time );

LIST *journal_account_name_list(
        full_name,
        street_address,
        transaction_date_time );

LIST *journal_binary_list(
        full_name,
        street_address,
        transaction_date_time,
        transaction_amount,
        debit_account,
        credit_account );

LIST *journal_year_list(
        int year,
        char *account_name,
        boolean fetch_memo );
```

TRANSACTION 1

```
char *full_name;
char *street_address;
char *transaction_date_time;
double transaction_amount;
char *memo;
int check_number;
boolean lock_transaction;
char *property_street_address;
char *program_name;
LIST *journal_list;

TRANSACTION *transaction_calloc(
        void );

TRANSACTION *transaction_new(
        full_name,
        street_address,
        transaction_date_time );

TRANSACTION *transaction_fetch(
        full_name,
        street_address,
        transaction_date_time );

char *transaction_primary_where(
        full_name,
        street_address,
        transaction_date_time );

void transaction_refresh(
        full_name,
        street_address,
        transaction_date_time,
        transaction_amount,
        memo,
        check_number,
        lock_transaction,
        journal_list );

/* Returns inserted transaction_date_time */
char *transaction_insert(
        full_name,
        street_address,
        transaction_date_time,
        transaction_amount,
        memo,
        check_number,
        lock_transaction );

FILE *transaction_insert_pipe_open(
        void );

/* Returns inserted transaction_date_time */
char *transaction_insert_pipe(
        insert_pipe,
        full_name,
        street_address,
        transaction_date_time,
        transaction_amount,
        memo,
        check_number,
        lock_transaction,
        journal_list );

/* Returns transaction_list */
LIST *transaction_list_insert(
        transaction_list );

void transaction_list_journal_insert(
        transaction_list );

void transaction_update(
        transaction_amount,
        full_name,
        street_address,
        transaction_date_time );

char *transaction_select(
        void );

char *transaction_primary_where(
        full_name,
        street_address,
        transaction_date_time );

TRANSACTION *transaction_parse(
        input_buffer );

/* Performs journal_propagate() */
void transaction_delete(
        full_name,
        street_address,
        transaction_date_time );

TRANSACTION *transaction_seek(
        transaction_list,
        full_name,
        street_address,
        transaction_date_time );

/* Returns race-free transaction_date_time */
char *transaction_race_free(
        transaction_date_time );
```



TRANSACTION 2

```
TRANSACTION *transaction_binary(
        full_name,
        street_address,
        transaction_date_time,
        double transaction_amount,
        char *memo,
        char *debit_account,
        char *credit_account );

char *transaction_memo(
        memo );

void transaction_check_insert(
        insert_pipe,
        check_number );

void transation_lock_insert(
        insert_pipe,
        lock_transaction );

boolean transaction_exists(
        transaction_date_time );

char *transaction_display(
        transaction );

char *transaction_escape_full_name(
        full_name );

TRANSACTION *transaction_check_seek(
        transaction_list,
        check_number );

boolean transaction_closing_entry_exists(
        char *transaction_date );

char *transaction_date_time_closing(
        char *transaction_date,
        boolean preclose_time,
        boolean transaction_closing_entry_exists() );
```

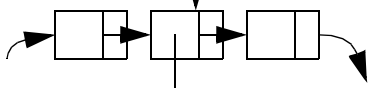
STATEMENT

```
/* Usage */
STATEMENT *statement_new(
        char *application_name,
        char *session,
        char *login_name,
        char *role_name,
        char *process_name,
        boolean exists_logo_filename,
        LIST *filter_element_name_list,
        char *as_of_date,
        int prior_year_count,
        LIST *fund_name_list,
        char *subclassification_option_string,
        char *fund_aggregation_string,
        char *output_medium_string );

STATEMENT *statement_steady_state(
        char *application_name,
        char *session,
        char *login_name,
        char *role_name,
        char *process_name,
        boolean exists_logo_filename,
        LIST *filter_element_name_list,
        char *as_of_date,
        int prior_year_count,
        LIST *fund_name_list,
        char *subclassification_option_string,
        char *fund_aggregation_string,
        char *output_medium_string,
        STATEMENT *statement );

enum subclassification_option
        statement_subclassification_option(
            subclassification_option_string );

enum fund_aggregation
        statement_fund_aggregation(
            fund_name_list,
            fund_aggregation_string );

enum output_medium
        statement_output_medium(
            output_medium_string );

LIST *statement_fund_name_list( void );

char *statement_begin_date_string(
        char *as_of_date );

char *statement_title(
        char *application_name,
        boolean exists_logo_filename,
        char *process_name );

char *statement_subtitle(
        statement_begin_date_string(),
        char *as_of_date,
        enum fund_aggregation );

LIST *statement_fund_list(
        application_name,
        session,
        login_name,
        role_name,
        filter_element_name_list,
        statement_begin_date_string(),
        as_of_date,
        prior_year_count,
        fund_name_list,
        char *progam_name,
        statement_subclassification_option(),
        statement_fund_aggregation() );

boolean statement_exists_closing_entry(
        char *as_of_date );
```

subclassification_aggregate,
subclassification_display,
subclassification_omit

single_fund,
sequential,
consolidated,
no_fund

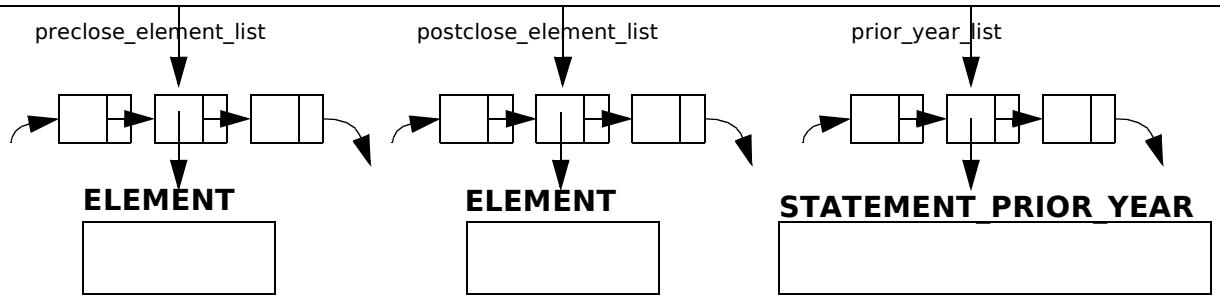
output_table,
output_PDF
output_spreadsheet,
output_stdout



STATEMENT_FUND

STATEMENT_FUND 1

```
/* Usage */
STATEMENT_FUND *statement_fund_fetch(
    char *application_name,
    char *session,
    char *login_name,
    char *role_name,
    LIST *filter_element_name_list,
    char *begin_date_string,
    char *as_of_date,
    int prior_year_count,
    char *fund_name,
    enum subclassification_option );

/* Process */
LIST *statement_fund_element_list(
        char *application_name,
        char *session,
        char *login_name,
        char *role_name,
        LIST *filter_element_name_list,
        char *begin_date_string,
        char *transaction_date_time,
        char *fund_name,
        enum subclassification_option );

LIST *preclose_element_list =
        statement_fund_element_list(
            application_name,
            session,
            login_name,
            role_name,
            filter_element_name_list,
            begin_date_string,
            transaction_date_time_closing(
                as_of_date,
                1 /* preclose_time_boolean */ ),
            fund_name,
            subclassification_option );

LIST *postclose_element_list =
        statement_fund_element_list(
            application_name,
            session,
            login_name,
            role_name,
            filter_element_name_list,
            begin_date_string,
            transaction_date_time_closing(
                as_of_date,
                0 /* not preclose_time_boolean */ ),
            fund_name,
            subclassification_option );

LIST *prior_year_list;

for years_ago = 1 to prior_year_count
{
        list_set(
            prior_year_list,
            statement_prior_year_new(
                application_name,
                session,
                login_name,
                role_name,
                filter_element_name_list,
                as_of_date,
                years_ago,
                fund_name,
                subclassification_option ) );
}

void statement_fund_list_steady_state(
        LIST *statement_fund_list );

STATEMENT_FUND *statement_fund_steady_state(
        LIST *preclose_element_list,
        LIST *postclose_element_list,
        int prior_year_count,
        STATEMENT_FUND *statement_fund );

LIST *statement_fund_steady_state_prior_year_list(
        int prior_year_count,
        LIST *preclose_element_list,
        STATEMENT_FUND *statement_fund );

double preclose_debit_total =
        element_list_debit_total(
            preclose_element_list );

double preclose_credit_total =
        element_list_credit_total(
            preclose_element_list );

double postclose_debit_total =
        element_list_debit_total(
            postclose_element_list );

double postclose_credit_total =
        element_list_credit_total(
            postclose_element_list );

boolean statement_fund_exists_postclose(
        LIST *statement_fund_list );

boolean  statement_fund_exists_prior_year(
        LIST *statement_fund_list );

STATEMENT_FUND *statement_fund_seek(
        char *fund_name,
        LIST *statement_fund_list );
```

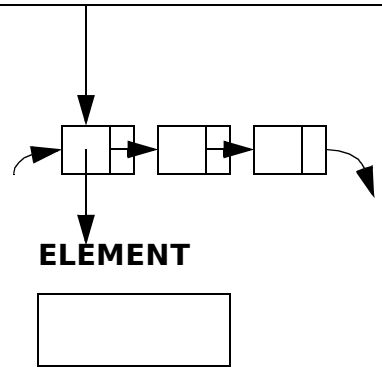


STATEMENT_FUND 2

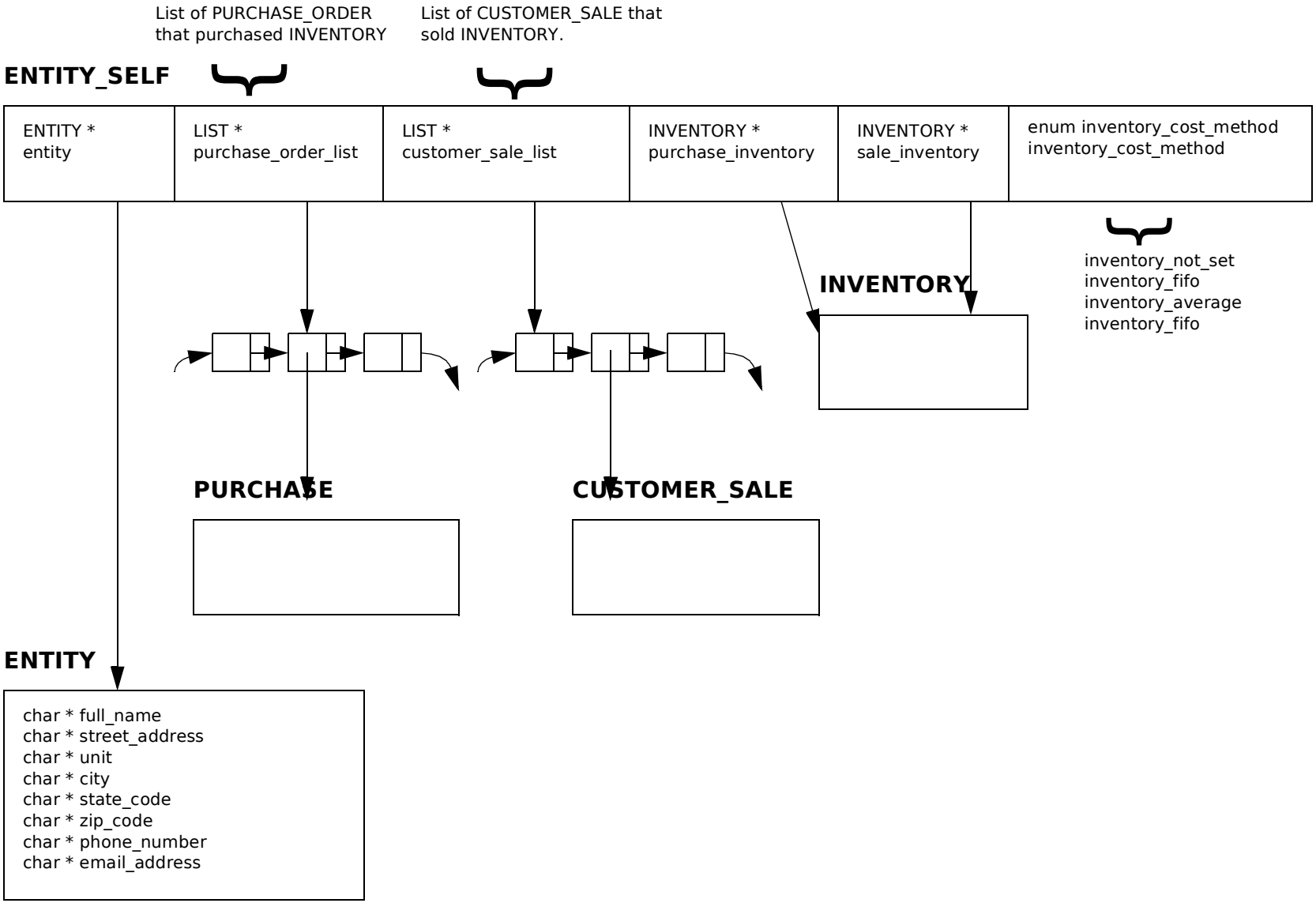
```
/* Process */
void statement_fund_list_net_income_set(
	LIST *statement_fund_list );

double statement_fund_total(
	LIST *subclassification_list,
	LIST *account_list );

double revenue_total =
		element_seek(
			ELEMENT_REVENUE,
			preclose_element_list )->
				element_balance_total() );

double expense_total =
		element_seek(
			ELEMENT_EXPENSE,
			preclose_element_list )->
				element_balance_total() );

double gain_total =
		element_seek(
			ELEMENT_GAIN,
			preclose_element_list )->
				element_balance_total() );

double loss_total =
		element_seek(
			ELEMENT_LOSS,
			preclose_element_list )->
				element_balance_total() );

double statement_fund_net_income(
		revenue_total,
		expense_total,
		gain_total,
		loss_total );

int statement_fund_net_income_percent(
		statement_fund_net_income(),
		revenue_total );
```

STATEMENT_PRIOR_YEAR

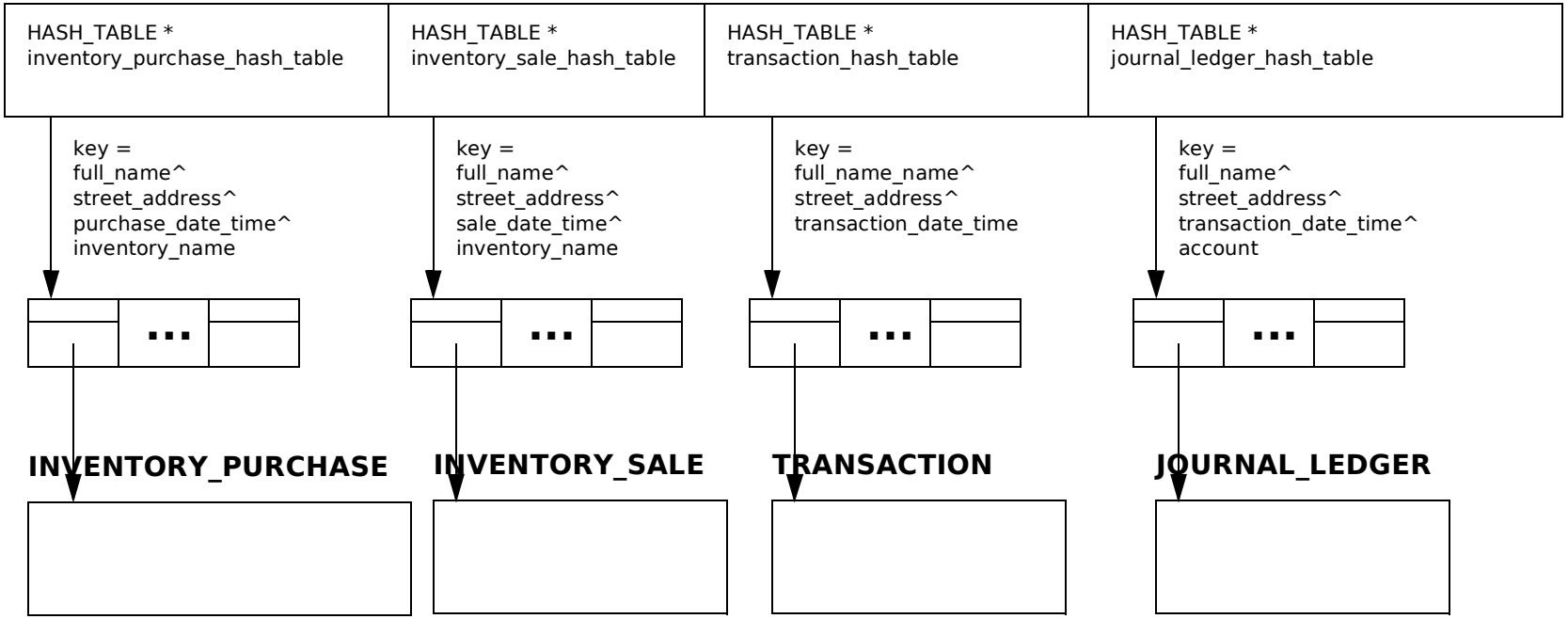
```
/* Usage */
STATEMENT_PRIOR_YEAR *
    statement_prior_year_new(
        char *application_name,
        char *session,
        char *login_name,
        char *role_name,
        LIST *filter_element_name_list,
        char *as_of_date,
        int years_ago,
        char *fund_name,
        enum subclassification_option );

/* Attributes */
char *application_name;
char *session;
char *login_name;
char *role_name;
LIST *filter_element_name_list;
char *as_of_date;
int years_ago;
char *fund_name;
enum subclassification_option subclassification_option;

/* Process */
char *prior_date_string =
    strdup(
        date_yyyy_mm_dd_display(
            date_subtract_year(
                date_yyyy_mm_dd_new(
                    as_of_date ),
                years_ago ) ) );

char *begin_date_string =
    statement_begin_date_string(
        prior_date_string );

LIST *prior_year_element_list =
    statement_fund_element_list(
        application_name,
        session,
        login_name,
        role_name,
        filter_element_name_list,
        begin_date_string,
        prior_date_string,
        fund_name,
        subclassification_option );

LIST *statement_prior_year_heading_list(
    LIST *prior_year_list );

LIST *statement_prior_year_delta_list(
    char *account_name,
    LIST *prior_year_list );
```



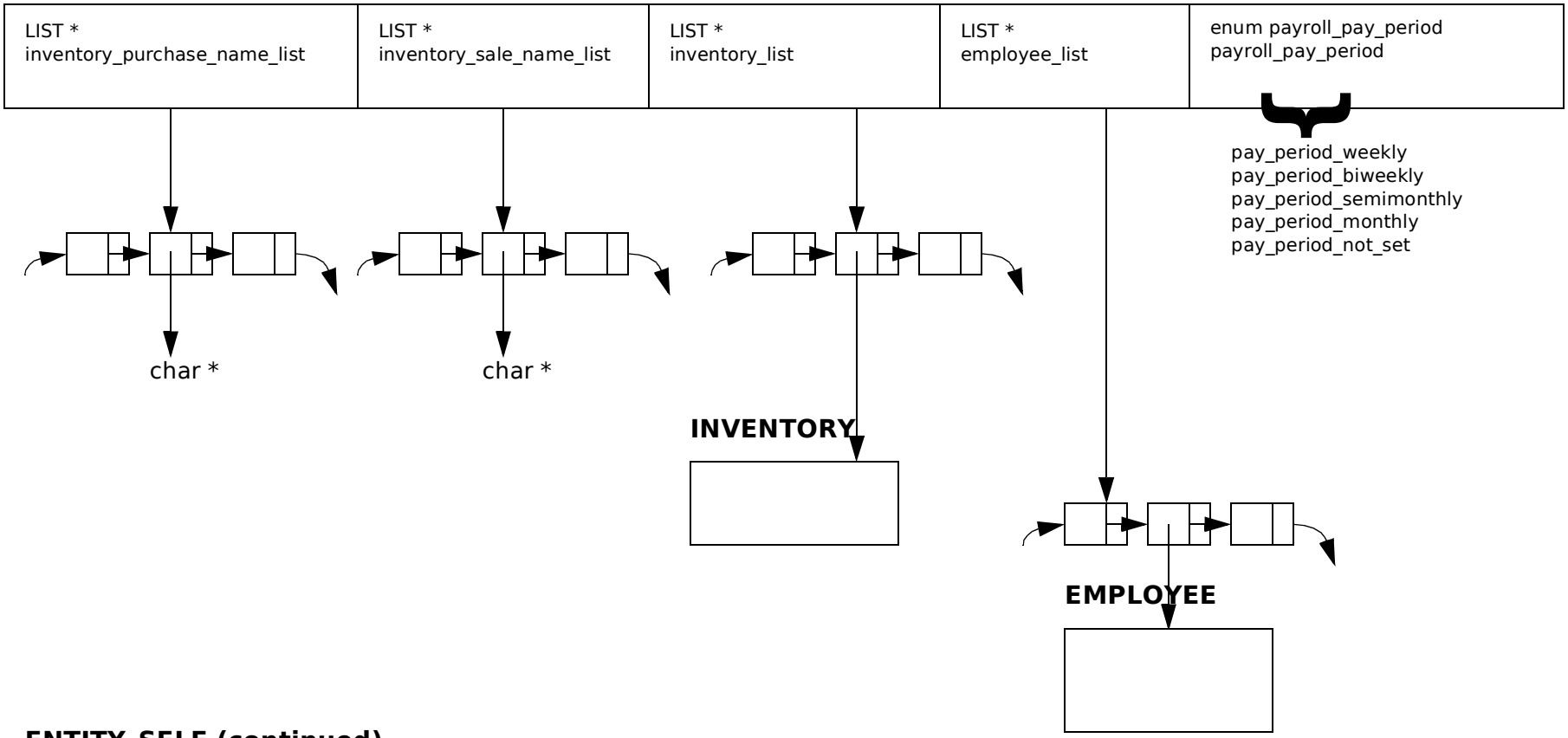
PredictiveBooks Data Structure



ENTITY_SELF (continued)



ENTITY_SELF (continued)



ENTITY_SELF (continued)

```
char *payroll_beginning_day
double social_security_combined_tax_rate
int social_security_payroll_ceiling
double medicare_combined_tax_rate
double medicare_additional_withholding_rate
int medicare_additional_gross_pay_floor
double federal_withholding_allowance_period_value
double federal_nonresident_withholding_income_premium
double state_withholding_allowance_period_value
double state_itemized_allowance_period_value
int federal_unemployment_wage_base
double federal_unemployment_tax_standard_rate
double federal_unemployment_threshold_rate
double federal_unemployment_tax_minimum_rate
int state_unemployment_wage_base
double state_unemployment_tax_rate
double state_sales_tax_rate;

char *entity_self_paypal_cash_account_name();
```

PredictiveBooks Data Structure

SALE (1)

```
ENTITY *customer_entity =
        entity_fetch(
            full_name,
            street_address );

char *sale_date_time;
char *completed_date_time;
double shipping_revenue;
enum title_passage_rule title_passage_rule;
double sales_tax_database;
char *transaction_date_time_database;

/* Returns sale_steady_state() */
SALE *sale_fetch(
        full_name,
        street_address,
        sale_date_time );

char *sale_primary_where(
        full_name,
        street_address,
        sale_date_time );

double sale_gross_revenue(
        inventory_sale_total(),
        fixed_service_sale_total(),
        hourly_service_sale_total() );

double sale_sales_tax(
        inventory_sale_list(),
        entity_self_sales_tax_rate() );

double sale_invoice_amount(
        sale_gross_revenue(),
        sale_sales_tax(),
        shipping_charge );

double Sale_amount_due(
        sale_invoice_amount(),
        customer_payment_total() )
{
        return A - B;
}

TRANSACTION *sale_transaction(
        full_name,
        street_address,
        completed_date_time,
        invoice_amount,
        gross_revenue,
        sale_tax,
        shipping_charge,
        account_receivable(),
        account_revenue(),
        account_shipping_revenue(),
        account_sales_tax_payable() );

void sale_update(
        inventory_sale_total(),
        fixed_service_sale_total(),
        hourly_service_sale_total(),
        sale_gross_revenue()
        sale_sales_tax(),
        sale_invoice_amount(),
        payment_total(),
        sale_amount_due(),
        sale_transaction()->
            transaction_date_time,
        full_name,
        street_address,
        sale_date_time );

SALE *sale_steady_state(
        full_name,
        street_address,
        sale_date_time,
        inventory_sale_list(),
        fixed_service_sale_list(),
        hourly_service_sale_list(),
        customer_payment_list(),
        shipping_charge,
        entity_self_sales_tax_rate(),
        sales_tax_database,
        completed_date_time,
        transaction_date_time_database.
        account_receivable(),
        account_revenue(),
        account_shipping_revenue(),
        account_sales_tax_payable() );
```

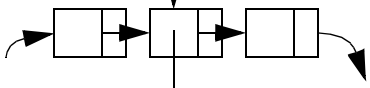
TRANSACTION

ENTITY

FOB_shipping
FOB_destination
title_passage_rule_null

SALE (2)

```
LIST *customer_payment_list(
        full_name,
        street_address,
        sale_date_time );
```



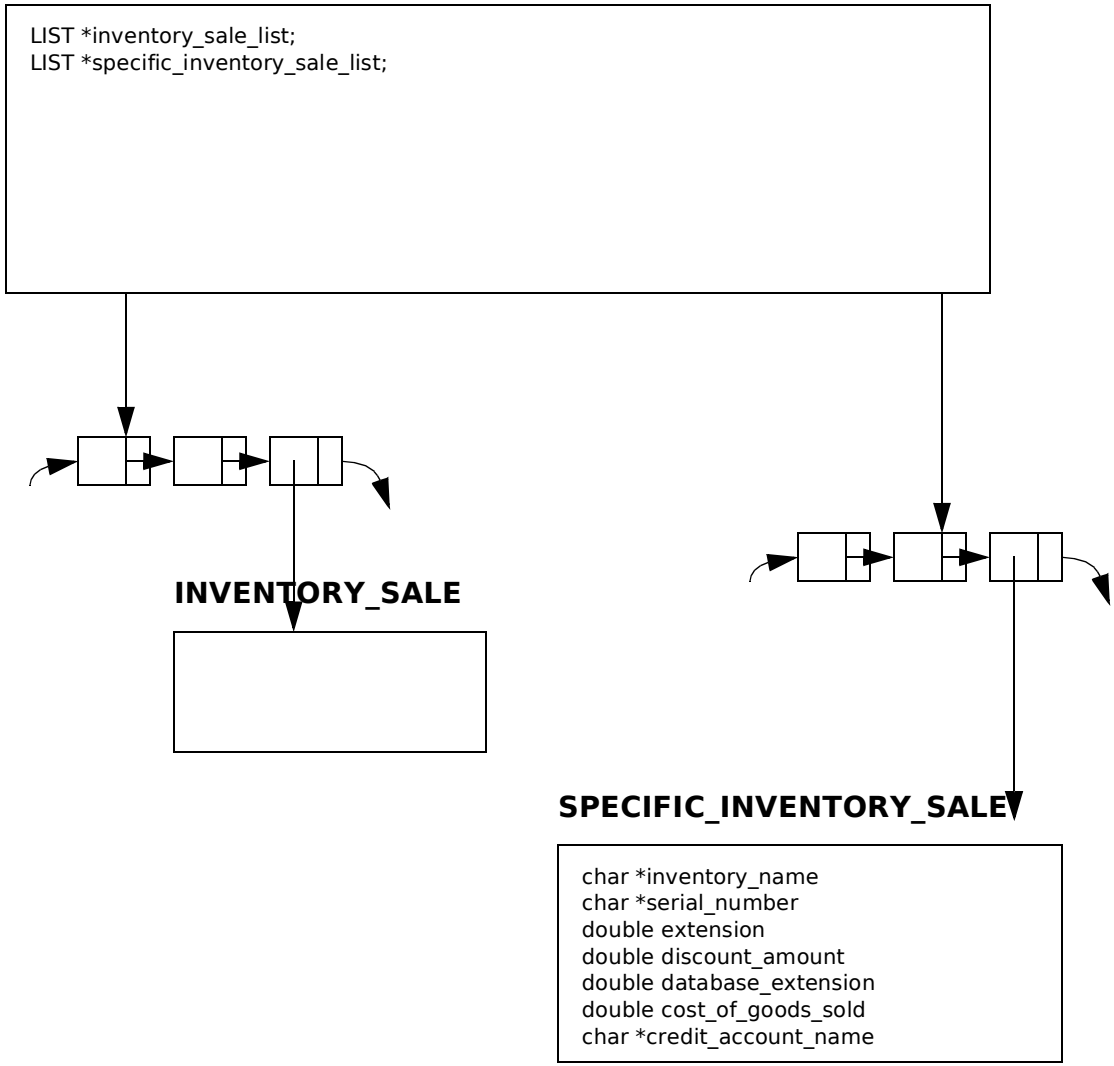
CUSTOMER_PAYMENT

```
ENTITY *customer_entity =
        entity_fetch(
            full_name,
            street_address );

char *sale_date_time;
char *payment_date_time;
double payment_amount;
int check_number;

double customer_payment_total(
        customer_payment_list() )
{
        return sum( A->payment_amount );
}

CUSTOMER_PAYMENT *customer_payment_new(
        full_name,
        street_address,
        sale_date_time,
        payment_date_time );

char *customer_payment_select(
        void );

char *customer_payment_sys_string(
        where );

LIST *customer_payment_system_list(
        sys_string );

CUSTOMER_PAYMENT *customer_payment_parse(
        input );

TRANSACTION *customer_payment_transaction(
        full_name,
        street_address,
        payment_date_time,
        payment_amount,
        account_cash(),
        account_receivable() );
```

ENTITY

TRANSACTION

SALE (3)


LIST *inventory_sale_list;
LIST *specific_inventory_sale_list;
INVENTORY_SALE
SPECIFIC_INVENTORY_SALE
char *inventory_name
char *serial_number
double extension
double discount_amount
double database_extension
double cost_of_goods_sold
char *credit_account_name

PredictiveBooks Data Structure

HOURLY_SERVICE_SALE

```
ENTITY *customer_entity =
        entity_new(
            full_name,
            street_address );
char *sale_date_time;
char *service_name;
char *service_description;
double hourly_rate;
double discount_amount;
double work_hours_database;
int estimated_hours;
double net_revenue_database;

char *hourly_service_sale_revenue_account_name(
        service_name,
        service_description );

LIST *hourly_service_work_list =
        work_hourly_list(
            full_name,
            street_address,
            sale_date_time,
            service_name,
            service_description );

double hourly_service_sale_work_hours =
        work_list_hours(
            hourly_service_work_list() );

double hourly_service_sale_net_revenue(
        hourly_rate,
        hourly_service_sale_work_hours,
        discount_amount )
{
        return (A x B) - C;
}
```

SALE (4)

```
LIST *    fixed_service_sale_list
LIST *    hourly_service_sale_list
```

FIXED_SERVICE_SALE

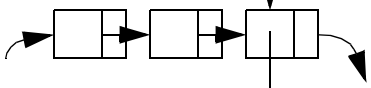
```
ENTITY *customer_entity =
        entity_new(
            full_name,
            street_address );
char *sale_date_time;
char *service_name;
double fixed_price;
double discount_amount;
double work_hours_database;
int estimated_hours;
double net_revenue_database;

char *fixed_service_sale_revenue_accouint_name(
        service_name );

LIST *fixed_service_sale_work_list =
        work_fixed_list(
            full_name,
            street_address,
            sale_date_time,
            service_name );

double fixed_service_sale_work_hours =
        work_list_hours(
            fixed_service_sale_work_list() );

double fixed_service_sale_net_revenue(
        fixed_price,
        discount_amount )
{
        return A - B;
}
```

WORK

```
ENTITY *customer_entity = entity_new();
char *sale_date_time;
char *service_name;
char *service_description;
char *begin_date_time;
char *end_date_time;
char *login_name;
char *activity_name;

double work_hours(
        begin_date_time,
        end_date_time );

double work_list_hours(
        work_list );
```

PURCHASE (1)

```
ENTITY *vendor_entity =
        entity_fetch(
            full_name,
            street_address );

char *purchase_date_time;
double sales_tax;
double freight_in;
char *arrived_date_time;
char *program_name;
char *property_street_address;

double purchase_equipment_total(
        purchase_equipment_list() )
{
        return sum(
                A->
                quipment_purchase->
                purchase_price );
}

double Purchase_invoice_amount(
        purchase_equipment_total(),
        sales_tax,
        freight_in );

double purchase_fetch_amount_due(
        full_name,
        street_address,
        purchase_date_time );

double Purchase_amount_due(
            Purchase_invoice_amount(),
            vendor_payment_total() )
{
        return A - B;
}

LIST *purchase_equipment_list(
        full_name,
        street_address,
        purchase_date_time );

TRANSACTION *purchase_transaction(
        full_name,
        street_address,
        arrived_date_time,
        Purchase_invoice_amount(),
        purchase_asset_account_name(
            purchase_equipment_list() ),
        account_payable() );

void purchase_update(
        purchase_equipment_total(),
        Purchase_invoice_amount(),
        vendor_payment_total(),
        Purchase_amount_due(),
        purchase_transaction()->
            transaction_date_time,
        full_name,
        street_address,
        purchase_date_time );

/* Returns true transaction_date_time */
char *purchase_transaction_refresh(
        purchase_transaction()->
            transaction_amount,
        purchase_transaction()->
            journal_list,
        purchase_transaction->memo,
        full_name,
        street_address,
        purchase_transaction()->
            transaction_date_time );

LIST *purchase_journal_list(
        Purchase_invoice_amount(),
        purchase_asset_account_name(),
        account_payable() );
```



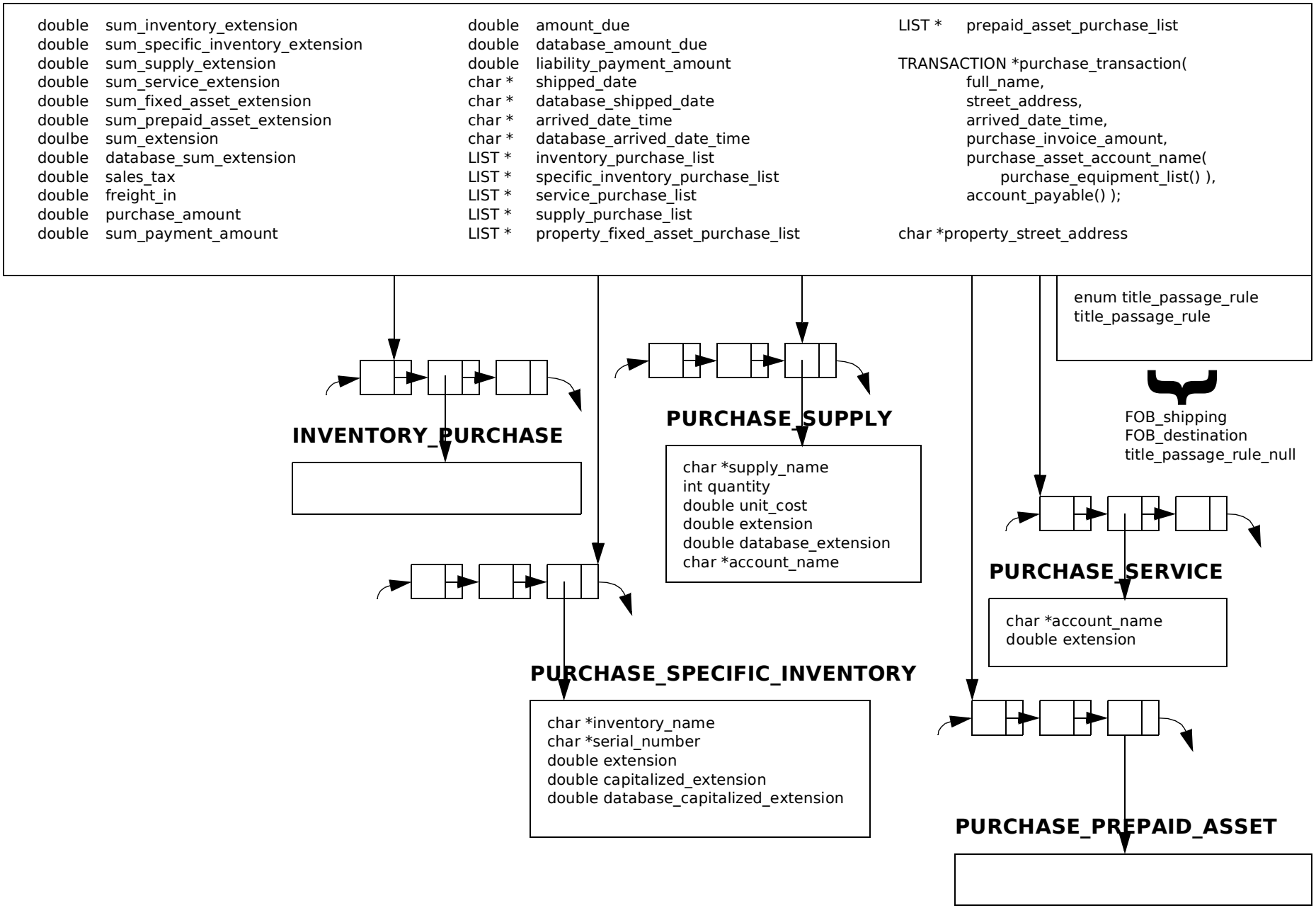
EQUIPMENT_PURCHASE

```
char *asset_name;
char *serial_number;
ENTITY *vendor_entity;
char *purchase_date_time;
char *service_placement_date;
double equipment_cost;
char *asset_account_name;
LIST *equipment_depreciation_list;
```

ENTITY

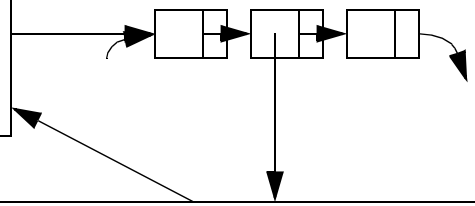
```
char *full_name;
char *street_address;
```

PURCHASE (2)


double sum_inventory_extension
double sum_specific_inventory_extension
double sum_supply_extension
double sum_service_extension
double sum_fixed_asset_extension
double sum_prepaid_asset_extension
doulbe sum_extension
double database_sum_extension
double sales_tax
double freight_in
double purchase_amount
double sum_payment_amount
double amount_due
double database_amount_due
double liability_payment_amount
char * shipped_date
char * database_shipped_date
char * arrived_date_time
char * database_arrived_date_time
LIST * inventory_purchase_list
LIST * specific_inventory_purchase_list
LIST * service_purchase_list
LIST * supply_purchase_list
LIST * property_fixed_asset_purchase_list
LIST * prepaid_asset_purchase_list
TRANSACTION *purchase_transaction(
full_name,
street_address,
arrived_date_time,
purchase_invoice_amount,
purchase_asset_account_name(
purchase_equipment_list()),
account_payable());
char *property_street_address
enum title_passage_rule
title_passage_rule
FOB_shipping
FOB_destination
title_passage_rule_null
INVENTORY_PURCHASE
PURCHASE_SUPPLY
char *supply_name
int quantity
double unit_cost
double extension
double database_extension
char *account_name
PURCHASE_SERVICE
char *account_name
double extension
PURCHASE_SPECIFIC_INVENTORY
char *inventory_name
char *serial_number
double extension
double capitalized_extension
double database_capitalized_extension
PURCHASE_PREPAID_ASSET

PURCHASE (3)

```
LIST *purchase_vendor_payment_list =
        vendor_payment_list(
            full_name,
            street_address,
            purchase_date_time );
```



VENDOR_PAYMENT

```
ENTITY *vendor_entity =
        entity_fetch(
            full_name,
            street_address );

char *purchase_date_time;
char *payment_date_time;
double payment_amount;
int check_number;

PURCHASE *purchase;

LIST *vendor_payment_list(
        full_name,
        street_address,
        purchase_date_time );

LIST *vendor_payment_fetch_list(
        where );

VENDOR_PAYMENT *vendor_payment_new(
        full_name,
        street_address,
        purchase_date_time,
        payment_date_time );

VENDOR_PAYMENT *vendor_payment_fetch(
        full_name,
        street_address,
        purchase_date_time,
        payment_date_time );

char *vendor_payment_select();

char *vendor_payment_primary_where(
        full_name,
        street_address,
        purchase_date_time,
        payment_date_time );

VENDOR_PAYMENT *vendor_payment_parse(
        input );

LIST *vendor_payment_system_list(
        sys_string );

double vendor_payment_total(
        LIST *vendor_payment_list() )
{
        return sum(
                A->
                vendor_payment->
                payment_amount );
}

TRANSACTION *vendor_payment_transaction(
        full_name,
        street_address,
        payment_date_time,
        payment_amount,
        check_number,
        account_payable(),
        account_cash(),
        account_uncleared_checks() );

LIST *vendor_payment_journal_list(
        payment_amount,
        check_number,
        account_payable(),
        account_cash(),
        account_uncleared_checks() );

void vendor_payment_insert(
        full_name,
        street_address,
        purchase_date_time,
        payment_date_time,
        payment_amount,
        check_number );

VENDOR_PAYMENT *vendor_payment_seek(
        vendor_payment_list,
        payment_date_time );

char *vendor_payment_update_sys_string(
        void );
```

EQUIPMENT_PURCHASE (1)

```
char *    asset_name;
char *    serial_number;
char *    service_placement_date;
double    equipment_cost;
int       estimated_useful_life_years;
int       estimated_useful_life_units;
int       estimated_residual_value;
int       declining_balance_n;
char *    depreciation_method;
double    tax_cost_basis;
char *    tax_recovery_period;
char *    disposal_date;
char *    equipment_account_name;
double    finance_accumulated_depreciation;
double    tax_accumulated_depreciation;
TAX_RECOVERY *tax_recovery;

ENTITY *vendor_entity =
          vendor_fetch(
               full_name,
               street_address );

EQUIPMENT_PURCHASE *
          equipment_purchase_new(
               asset_name,
               serial_number,
               full_name,
               street_address,
               purchase_date_time );

EQUIPMENT_PURCHASE *
          equipment_purchase_fetch(
               asset_name,
               serial_number,
               full_name,
               street_address,
               purchase_date_time );

EQUIPMENT_PURCHASE *
          equipment_purchase_parse(
               input );

char *equipment_purchase_select(
          void );

FILE *equipment_purchase_update_open(
          void );

void equipment_purchase_update(
          finance_accumulated_depreciation,
          tax_accumulated_depreciation,
          asset_name,
          serial_number,
          full_name,
          street_address,
          purchase_date_time );

LIST *equipment_purchase_list(
          full_name,
          street_address,
          purchase_date_time );
```

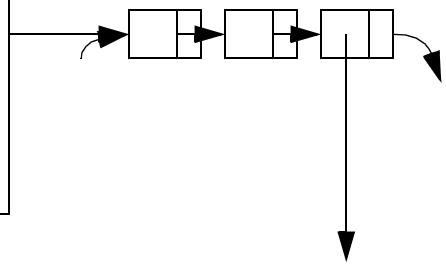
TAX_RECOVERY

char *	service_placecment_date	char *	disposal_date
double	tax_cost_basis	int	tax_year
char *	tax_recovery_period	double	recovery_amount
double	tax_recovery_period_years	double	recovery_percent

ENTITY

EQUIPMENT_PURCHASE (2)

```
LIST *depreciation_list =
        depreciation_list(
            asset_name,
            serial_number,
            full_name,
            street_address,
            purchase_date_time );

DEPRECIATION *equipment_purchase_depreciation(
        equipment_purchase,
        depreciation_date,
        prior_depreciation_date,
            transaction_date_time,
            units_produced );

LIST *equipment_purchase_depreciation_list(
        equipment_purchase_list );

/* Returns equipment_purchase_list with */
/* equipment_purchase->depreciation_list set. */
LIST *equipment_purchase_list_depreciate(
        equipment_purchase_list(),
        depreciation_date,
        boolean set_depreciation_date );
```



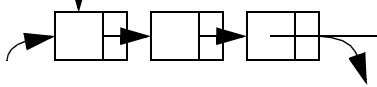
DEPRECIATION

```
char *asset_name;
char *serial_number;

ENTITY *vendor_entity =
        entity_new(
            full_name,
            street_address ) );

char *purchase_date_time;
char *depreciation_date;

double depreciation_accumulated_depreciation(
        prior_accumulated_depreciation,
        depreciation_amount() );

double depreciation_amount(
        depreciation_method,
        equipment_cost,
        estimated_residual_value,
        estimated_useful_life_years,
        estimated_useful_life_units,
        declining_balance_n,
        prior_depreciation_date,
        depreciation_date_string,
        finance_accumulated_depreciation,
        service_placement_date,
        units_produced );

double depreciation_straight_line(
        equipment_cost,
        estimated_residual_value,
        estimated_useful_life_years,
        prior_depreciation_date,
        depreciation_date,
        finance_accumulated_depreciation );

double depreciation_sum_of_years_digits(
        equipment_cost,
        estimated_residual_value,
        estimated_useful_life_years,
        prior_depreciation_date,
        depreciation_date,
        finance_accumulated_depreciation,
        service_placement_date );

double depreciation_double_declining_balance(
        equipment_cost,
        estimated_residual_value,
        estimated_useful_life_years,
        prior_depreciation_date,
        depreciation_date,
        finance_accumulated_depreciation );

double depreciation_n_declining_balance(
        equipment_cost,
        estimated_residual_value,
        estimated_useful_life_years,
        prior_depreciation_date,
        depreciation_date,
        finance_accumulated_depreciation,
        n );

double depreciation_units_of_production(
        equipment_cost,
        estimated_residual_value,
        estimated_useful_life_units,
        units_produced,
        finance_accumulated_depreciation );

double depreciation_fraction_of_year(
        prior_depreciation_date,
        depreciation_date );

double depreciation_amount_total(
        depreciation_list() );

LIST *depreciation_list(
        asset_name,
        serial_number,
        full_name,
        street_address,
        purchase_date_time );

TRANSACTION *depreciation_transaction(
        full_name,
        street_address,
        transaction_date_time,
        depreciation_amount,
        account_depreciation_expense(),
        account_accumulated_depreciation() );

LIST *depreciation_transaction_list(
        depreciation_list );
```

PURCHASE_PREPAID_ASSET

char *	full_name	char *	asset_account_name	double	accumulated_accrual
char *	street_address	char *	expense_account_name		
char *	purchase_date_time	double	extension		
char *	asset_name	double	accrual_period_years		

LIST *
accrual_list



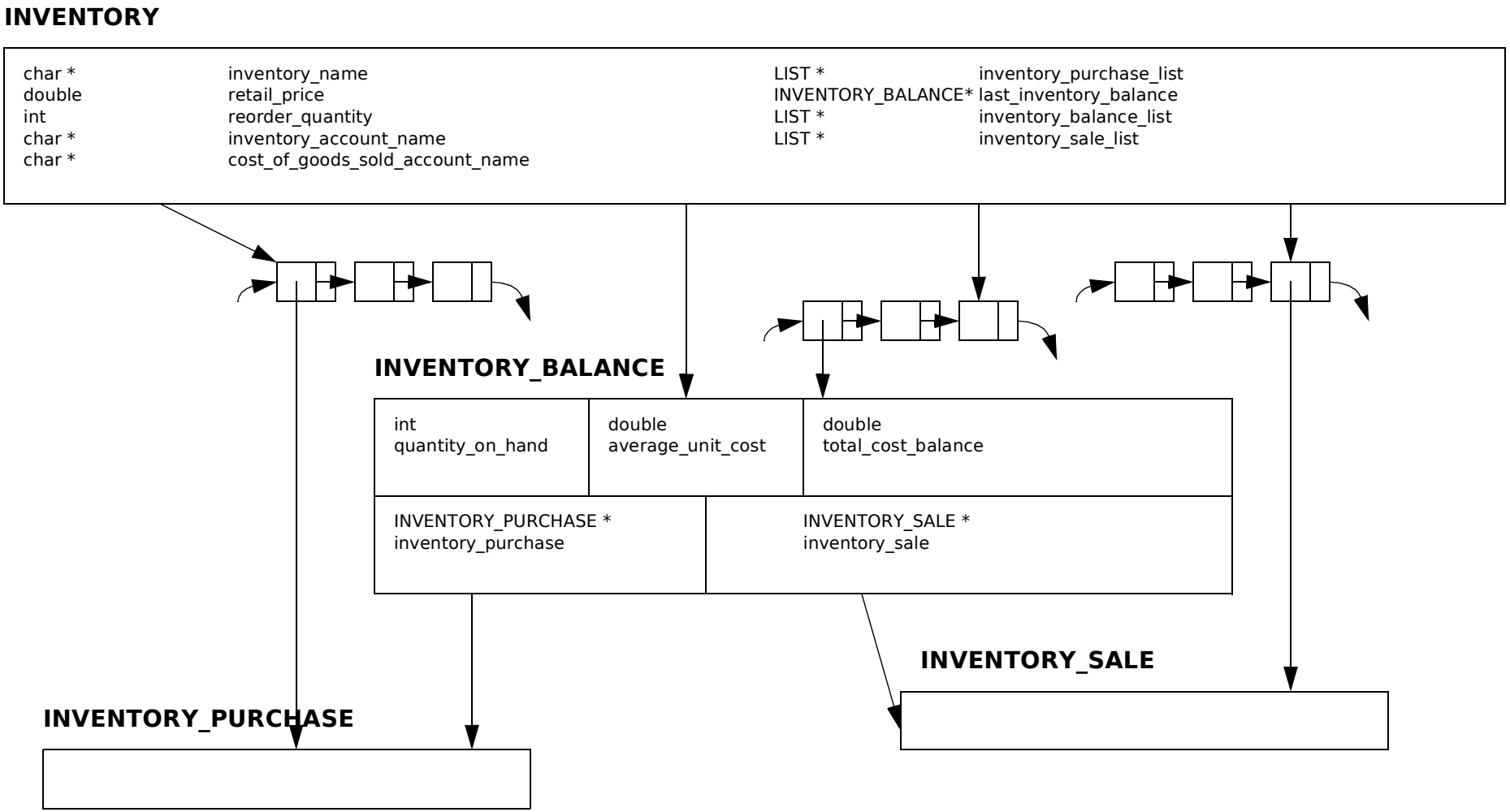
ACCRUAL

char *	full_name	double	accrual_amount
char *	street_address	double	database_accrual_amount
char *	purchase_date_time	char *	transaction_date_time
char *	asset_name	char *	database_transaction_date_time
char *	accrual_date	TRANSACTION *transaction	

VENDOR_PAYMENT

char * payment_date_time	double payment_amount	char * transaction_date_time	char * database_transaction_date_time	TRANSACTION * transaction

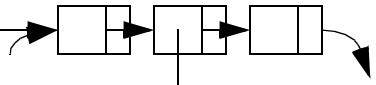
TRANSACTION


INVENTORY
char * inventory_name
double retail_price
int reorder_quantity
char * inventory_account_name
char * cost_of_goods_sold_account_name
LIST * inventory_purchase_list
INVENTORY_BALANCE* last_inventory_balance
LIST * inventory_balance_list
LIST * inventory_sale_list
INVENTORY_BALANCE
int quantity_on_hand
double average_unit_cost
double total_cost_balance
INVENTORY_PURCHASE * inventory_purchase
INVENTORY_SALE * inventory_sale
INVENTORY_SALE
INVENTORY_PURCHASE

PredictiveBooks Data Structure

INVENTORY_PURCHASE

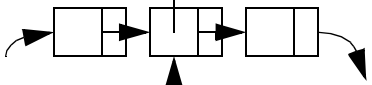
char *	full_name	int	database_arrived_quantity
char *	street_address	int	missing_quantity
char *	purchase_date_time	int	quantity_on_hand
char *	inventory_name	int	database_quantity_on_hand
int	ordered_quantity	int	layer_consumed_quantity
double	unit_cost	double	average_unit_cost
double	capitalized_unit_cost	double	database_average_unit_cost
double	database_capitalized_unit_cost	char *	arrived_date_time
double	extension	char *	inventory_account_name
double	database_extension	LIST *	inventory_purchase_return_list
int	arrived_quantity		



INVENTORY_PURCHASE_RETURN

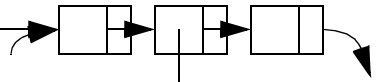
char *	return_date_time
int	returned_quantity
double	sales_tax
char *	transaction_date_time
char *	database_transaction_date_time
TRANSACTION *	transaction

TRANSACTION



INVENTORY_SALE

char *	full_name	double	database_extension
char *	street_address	double	cost_of_goods_sold
char *	sale_date_time	double	database_cost_of_goods_sold
char *	inventory_name	char *	completed_date_time
int	sold_quantity	char *	inventory_account_name
double	retail_price	char *	cost_of_goods_sold_account_name
double	discount_amount	LIST *	inventory_sale_return_list
double	extension	LIST *	layer_inventory_purchase_list



INVENTORY_SALE_RETURN

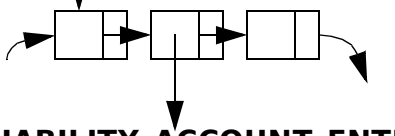
char *	return_date_time
int	returned_quantity
char *	transaction_date_time
char *	database_transaction_date_time
TRANSACTION *	transaction

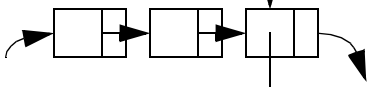
TRANSACTION

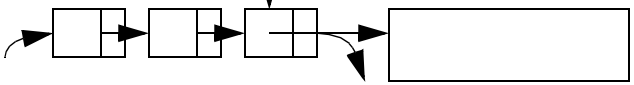
PredictiveBooks Data Structure

LIABILITY

```
/* Input */
int starting_check_number;
LIST *liability_account_entity_list();
LIST *liability_current_account_list();

LIST *input_entity_list =
        entity_full_street_list(
            full_name_list,
            street_address_list );

/* Process */

LIST *liability_tax_redirect_account_list(
        liability_current_account_list(),
        liability_account_entity_list() );

LIST *liability_entity_list(
        liability_tax_redirect_account_list()
            /* liability_account_list */,
        input_entity_list,
        dialog_box_payment_amount );

LIST *liability_after_balance_zero_entity_list(
        liability_entity_list(),
        liability_tax_redirect_account_list()
            /* liability_account_list */ );

LIST *liability_steady_state_entity_list(
        liability_after_balance_zero_entity_list() );

LIST *liability_transaction_list(
        liability_entity_list_steady_state(),
        liability_credit_account_name(
            starting_check_number ),
        account_loss() );
```



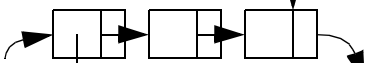


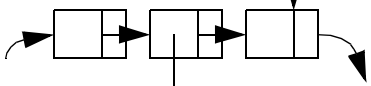


TRANSACTION

LIABILITY_ACCOUNT_ENTITY

```
char *account_name;
ENTITY *entity;
```



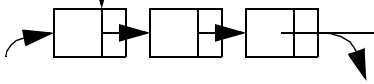


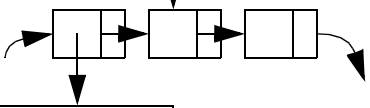
ENTITY

```
char *full_name;
char *street_address;
int check_number;
double dialog_box_payment_amount;

LIST *liability_after_balance_zero_journal_list(
        liability_account_list(),
        full_name,
        street_address );

char *liability_entity_debit_account_name(
        account->account_name );

double liability_entity_amount_due(
        liability_entity_journal_list() );

double liability_entity_payment_amount(
        dialog_box_payment_amount,
        liability_entity_amount_due() );

double liability_entity_loss_amount(
        dialog_box_payment_amount,
        liability_entity_payment_amount() )
{
        return A - B;
}

ENTITY *liability_steady_state_entity(
        entity,
        liability_entity_journal_list() );
```

ACCOUNT

```
char *account_name;

LIST *transaction_after_balance_zero_journal_list(
        account_name );
```





JOURNAL

```
char *account_name;
char *full_name;
char *street_address;
double debit_amount;
double credit_amount;
```

PredictiveBooks Data Structure

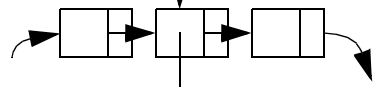
EMPLOYEE

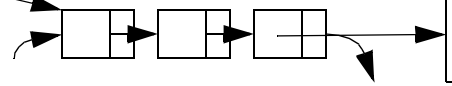
char *full_name
char *street_address
double hourly_wage
double period_salary
double commission_sum_extension_percent
double gross_pay_year_to_date
double database_gross_pay_year_to_date
double net_pay_year_to_date
double database_net_pay_year_to_date
char * federal_marital_status
int federal_withholding_allowances
int federal_withholding_additional_period_amount
boolean state_withholding_exempt
char *state_marital_status
int state_withholding_allowances
int state_itemized_deduction_allowances
int retirement_contribution_plan_employee_period_amount
int retirement_contribution_plan_employer_period_amount
int health_insurance_employee_period_amount
int health_insurance_employer_period_amount
double union_dues_period_amount
char *terminated_date

LIST *
employee_work_day_list

LIST *
employee_work_period_list

LIST *
customer_sale_list





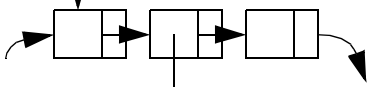
CUSTOMER_SALE

EMPLOYEE_WORK_PERIOD

char *full_name
char *street_address
int payroll_year
int payroll_period_number
char *begin_work_date
char *end_work_date
double regular_work_hours
double overtime_work_hours
double commission_sum_extension
double gross_pay
double net_pay
double payroll_tax_amount
double federal_tax_withholding_amount
double state_tax_withholding_amount
double social_security_employee_tax_amount
double social_security_employer_tax_amount
double medicare_employee_tax_amout
double medicare_employer_tax_amount
int
retirement_contribution_plan_employee_amount
int retirement_contribution_plan_employer_amount
int health_insurance_employee_amount
int health_insurance_employer_amount
double federal_unemployment_tax_amount
double state_unemployment_tax_amount
int union_dues_amount
char *transaction_date_time

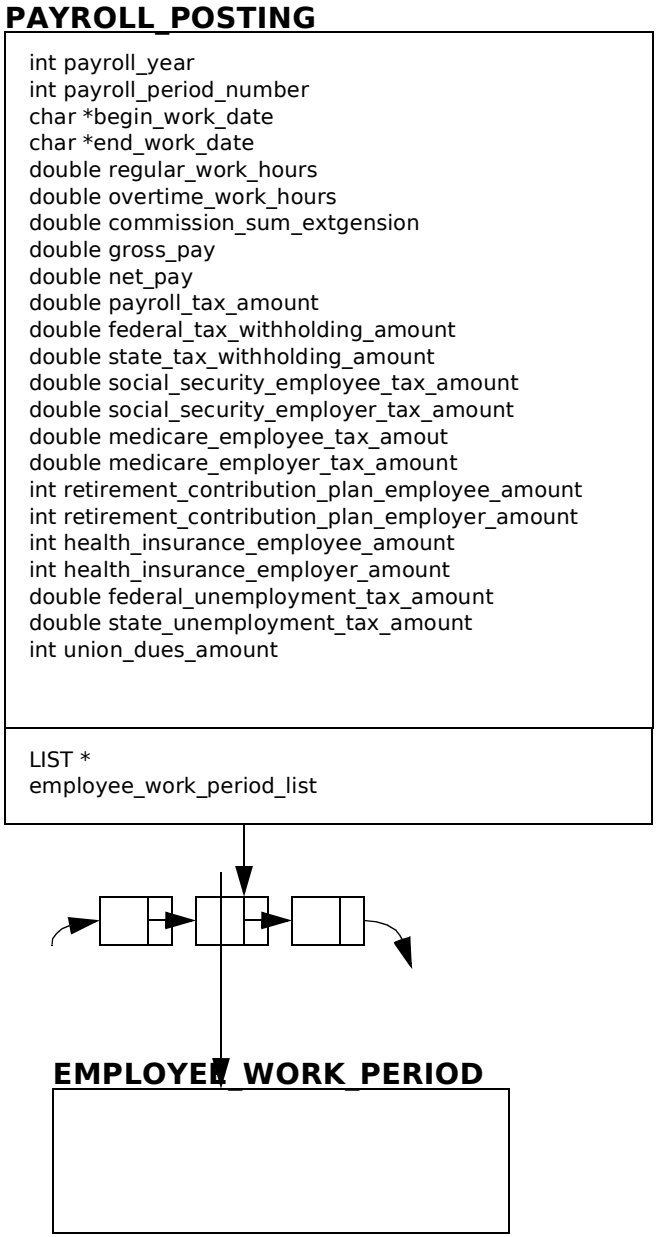
TRANSACTION *
transaction

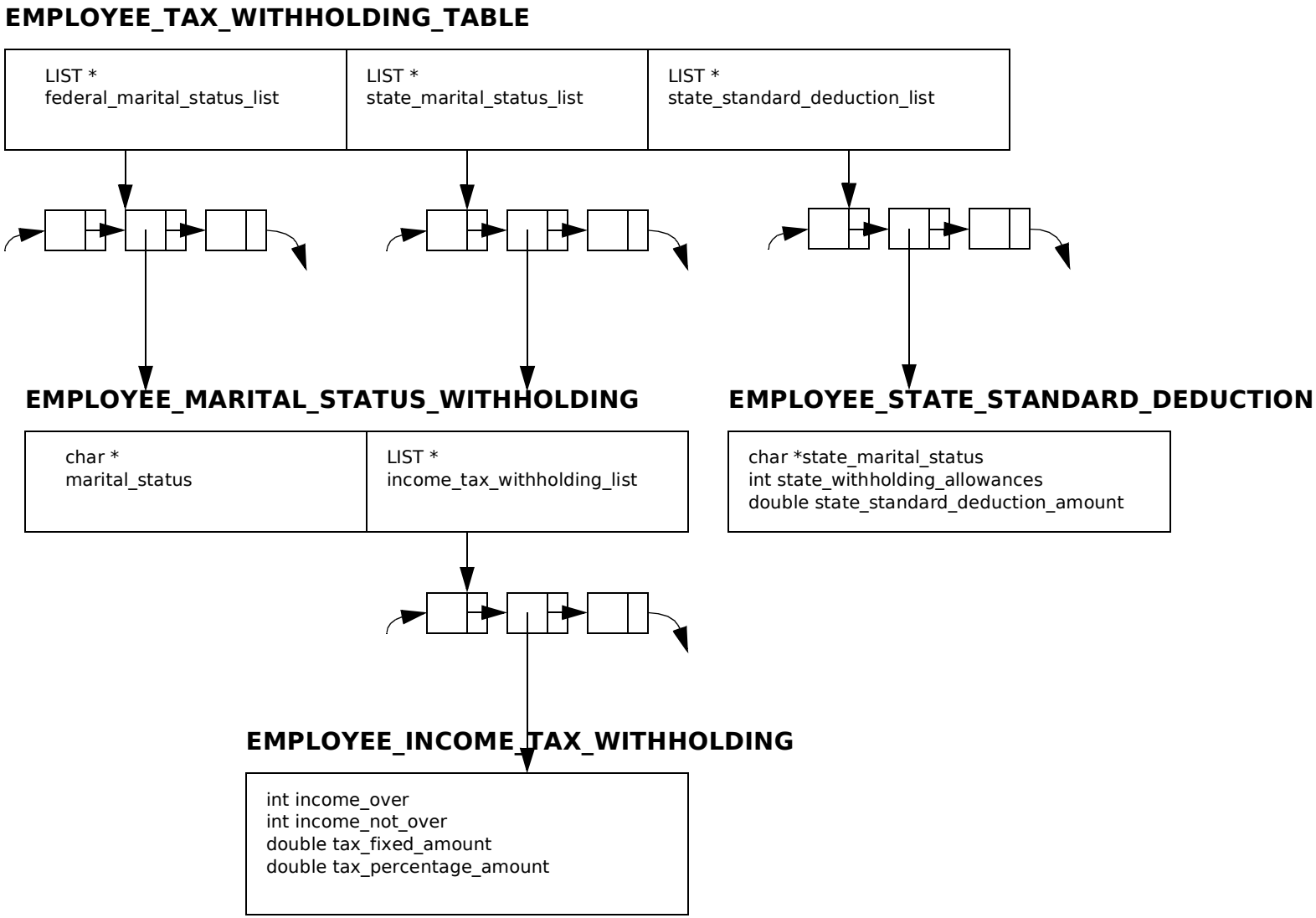
TRANSACTION



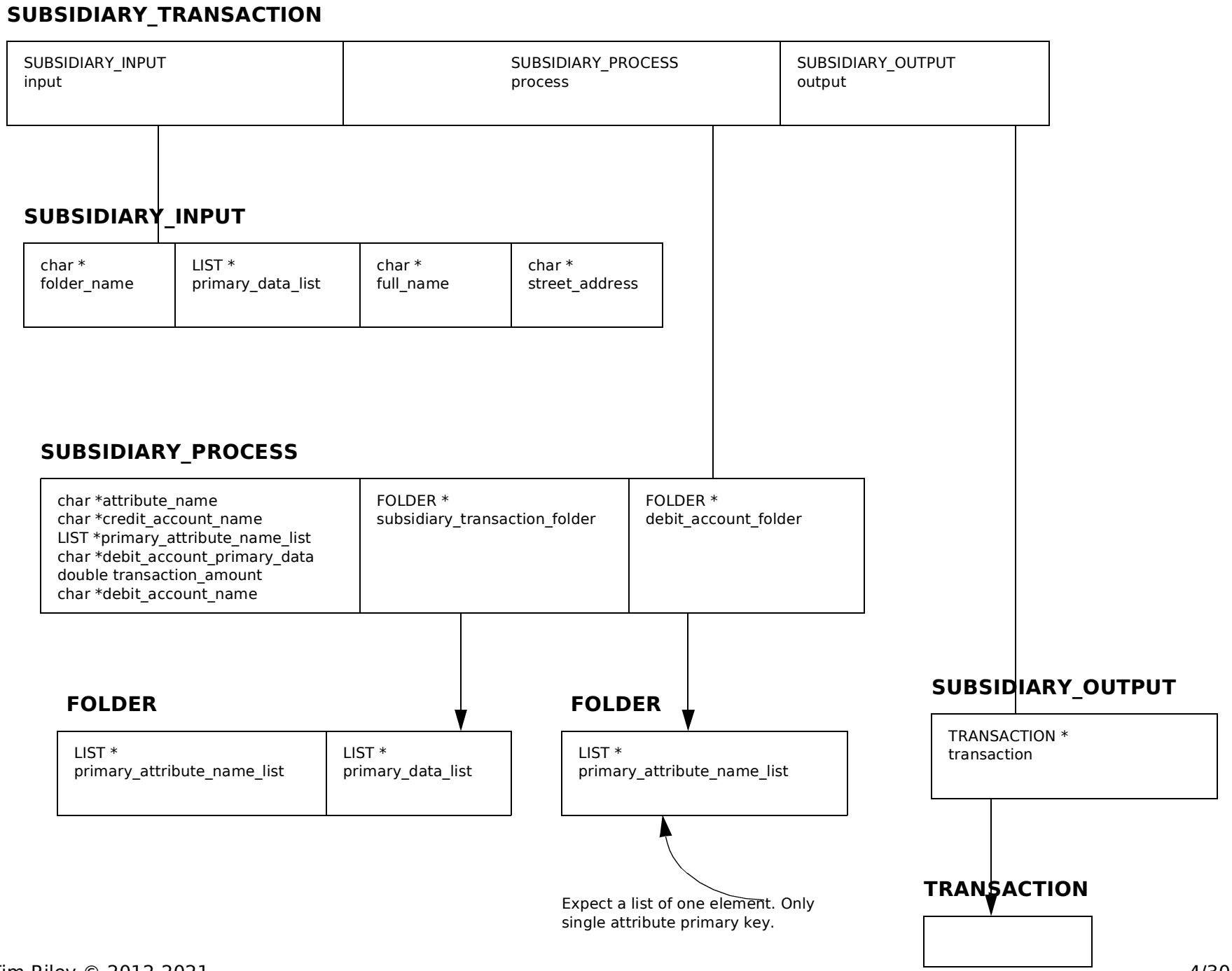
EMPLOYEE_WORK_DAY

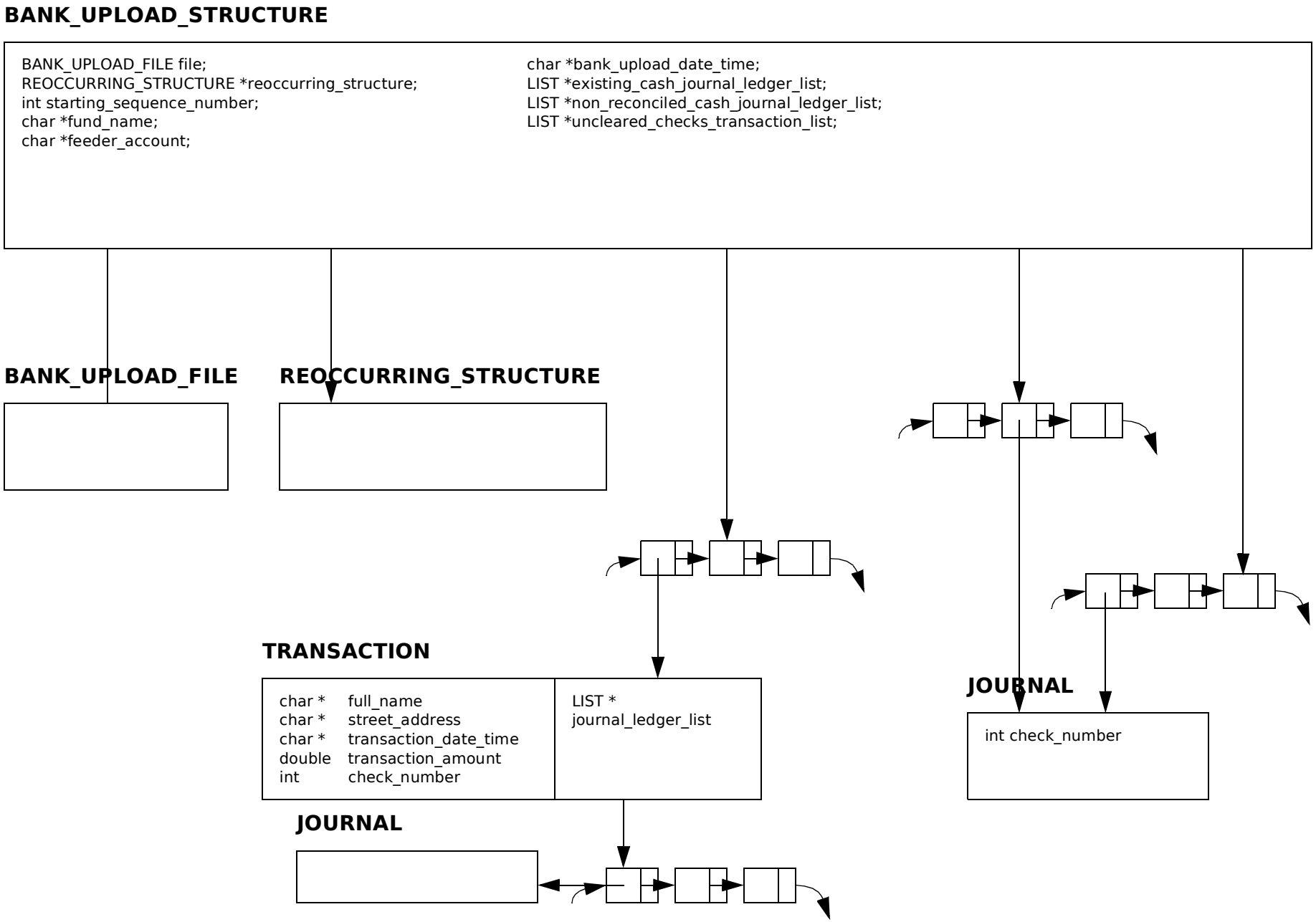
char *begin_work_date_time
char *end_work_date_time
double employee_work_hours
double database_employee_work_hours
boolean overtime_work_day

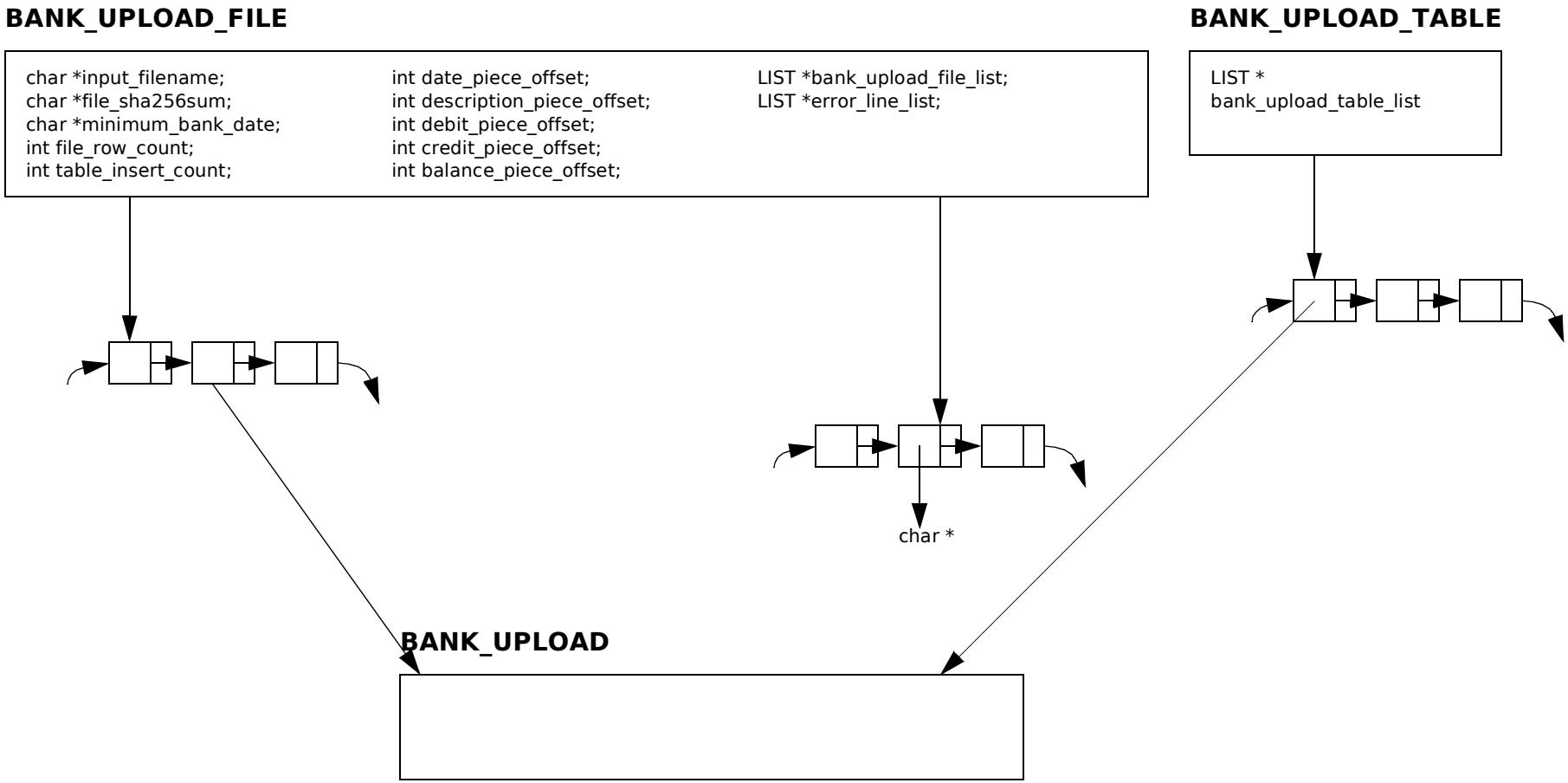

PAYROLL_POSTING
int payroll_year
int payroll_period_number
char *begin_work_date
char *end_work_date
double regular_work_hours
double overtime_work_hours
double commission_sum_extgension
double gross_pay
double net_pay
double payroll_tax_amount
double federal_tax_withholding_amount
double state_tax_withholding_amount
double social_security_employee_tax_amount
double social_security_employer_tax_amount
double medicare_employee_tax_amout
double medicare_employer_tax_amount
int retirement_contribution_plan_employee_amount
int retirement_contribution_plan_employer_amount
int health_insurance_employee_amount
int health_insurance_employer_amount
double federal_unemployment_tax_amount
double state_unemployment_tax_amount
int union_dues_amount
LIST *
employee_work_period_list
EMPLOYEE_WORK_PERIOD


EMPLOYEE_TAX_WITHHOLDING_TABLE
LIST *
federal_marital_status_list
LIST *
state_marital_status_list
LIST *
state_standard_deduction_list
EMPLOYEE_MARITAL_STATUS_WITHHOLDING
char *
marital_status
LIST *
income_tax_withholding_list
EMPLOYEE_STATE_STANDARD_DEDUCTION
char *state_marital_status
int state_withholding_allowances
double state_standard_deduction_amount
EMPLOYEE_INCOME_TAX_WITHHOLDING
int income_over
int income_not_over
double tax_fixed_amount
double tax_percentage_amount

PredictiveBooks Data Structure

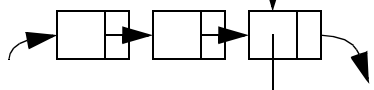



BANK_UPLOAD_STRUCTURE
BANK_UPLOAD_FILE file;
REOCCURRING_STRUCTURE *reoccurring_structure;
int starting_sequence_number;
char *fund_name;
char *feeder_account;
char *bank_upload_date_time;
LIST *existing_cash_journal_ledger_list;
LIST *non_reconciled_cash_journal_ledger_list;
LIST *uncleared_checks_transaction_list;
BANK_UPLOAD_FILE
REOCCURRING_STRUCTURE
TRANSACTION
char * full_name
char * street_address
char * transaction_date_time
double transaction_amount
int check_number
LIST *
journal_ledger_list
JOURNAL
int check_number
JOURNAL


BANK_UPLOAD_FILE
char *input_filename;
char *file_sha256sum;
char *minimum_bank_date;
int file_row_count;
int table_insert_count;
int date_piece_offset;
int description_piece_offset;
int debit_piece_offset;
int credit_piece_offset;
int balance_piece_offset;
LIST *bank_upload_file_list;
LIST *error_line_list;
BANK_UPLOAD_TABLE
LIST *
bank_upload_table_list
char *
BANK_UPLOAD

BANK_UPLOAD

char *bank_date char *bank_description int sequence_number double bank_amount double bank_running_balance	char *fund_name int check_number boolean existing_bank_upload boolean existing_transaction	LIST * reconciled_transaction_list	JOURNAL_LEDGER * cleared_journal_ledger

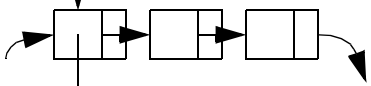


TRANSACTION

JOURNAL_LEDGER

BANK_UPLOAD

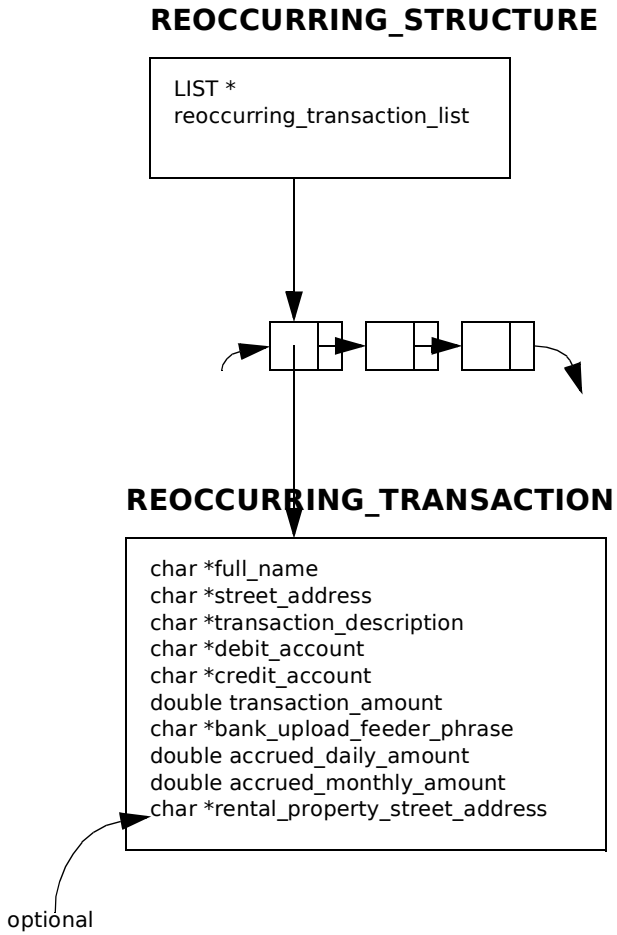
JOURNAL * feeder_check_number_existing_journal(LIST *existing_cash_journal_list, int check_number)	LIST * feeder_match_sum_existing_journal_list(LIST *existing_cash_journal_list, double abs_bank_amount, boolean check_debit)	TRANSACTION * feeder_phrase_match_build_transaction(LIST *reoccurring_transaction_list, char *bank_date, char *bank_description_embedded, double abs_bank_amount)



JOURNAL

JOURNAL

TRANSACTION


REOCCURRING_STRUCTURE
LIST *
reoccurring_transaction_list
REOCCURRING_TRANSACTION
char *full_name
char *street_address
char *transaction_description
char *debit_account
char *credit_account
double transaction_amount
char *bank_upload_feeder_phrase
double accrued_daily_amount
double accrued_monthly_amount
char *rental_property_street_address
optional

PAYPAL

```
/* Usage */
PAYPAL *paypal_fetch(
        spreadsheet_filename,
        date_label );

/* Process */
SPREADSHEET *spreadsheet =
        spreadsheet_fetch(
            spreadsheet_filename,
            date_label );
```

SPREADSHEET

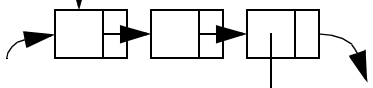
PAYPAL_DATASET

```
/* Input */
char *date_A;
char *time_B;
char *timezone_C;
char *full_name_D;
char *transaction_type_E;
char *transaction_status_F;
char *currency_G;
char *gross_revenue_H;
char *transaction_fee_I;
char *net_revenue_J;
char *from_email_address_K;
char *to_email_address_L;
char *transaction_ID_M;
char *shipping_address_N;
char *address_status_O;
char *item_title_P;
char *item_ID_Q;
char *shipping_handling_amount_R;
char *insurance_amount_S;
char *sales_tax_T;
char *option_1_name_U;
char *option_1_value_V;
char *option_2_name_W;
char *option_2_value_X;
char *reference_txn_ID_Y;
char *invoice_number_Z;
char *custom_number_AA;
char *quantity_AB;
char *receipt_ID_AC;
char *balance_AD;
char *address_line_1_AE;
char *address_2_district_neighborhood_AF;
char *town_city_AG;
char *state_province_AH;
char *zip_code_AI;
char *country_AJ;
char *contact_phone_number_AK;
char *subject_AL;
char *note_AM;
char *country_code_AN;
char *balance_impact_AO;
```

PredictiveBooks Data Structure

SPREADSHEET

```
/* Input */
char *spreadsheet_filename;
char *date_label;

/* Process */
LIST *spreadsheet_column_list(
        spreadsheet_filename,
        date_label );

char *maximum_date;

char *spreadsheet_minimum_date(
        &maximum_date,
        spreadsheet_filename );

char *spreadsheet_heading_data(
        spreadsheet_column_list(),
        input_row,
        heading );
```



SPREADSHEET_COLUMN

```
/* Usage */
SPREADSHEET_COLUMN *spreadsheet_column_new(
        heading,
        piece_offset );

/* Process */
SPREADSHEET_COLUMN *spreadsheet_column_heading_seek(
        spreadsheet_column_list,
        heading );
```

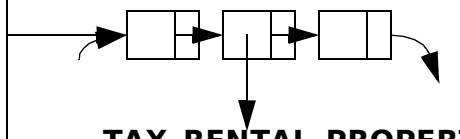
SPREADSHEET_UPLOAD_EVENT

```
/* Input */
char *spreadsheet_filename;

/* Process */
char *spreadsheet_upload_date_time =
        date_time19_now();

char *spreadsheet_upload_file_sha256sum(
        spreadsheet_filename );
```

TAX

```
/* Usage */
TAX *tax_new(
        char *tax_form_string,
        int tax_year );

/* Input */
char *tax_form_string;
int tax_year;

/* Attributes */
```

```
TAX_FORM *tax_form =
        tax_form_fetch();

TAX_RECOVERY *tax_recovery;

LIST *tax_rental_property_list =
        tax_rental_property_list_fetch(
            tax_form_string,
            tax_year );
```



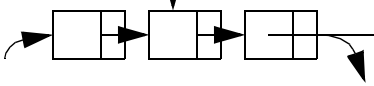
TAX_RENTAL_PROPERTY

TAX_FORM

```
/* Usage */
TAX_FORM *tax_form_fetch(
        char *tax_form_string,
        int tax_year,
        boolean fetch_line_list,
        boolean fetch_account_list,
        boolean fetch_journal_list );

/* Attributes */
char *tax_form_string;

/* Input */
int tax_year;

/* Process */
LIST *tax_form_line_list(
        tax_form_string,
        tax_year,
        fetch_account_list,
        fetch_journal_list );
```

TAX_RECOVERY

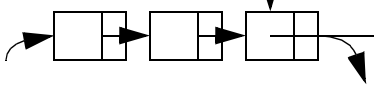
```
double fixed_asset_recovery_amount;
double prior_fixed_asset_recovery_amount;
double property_recovery_amount;
double prior_property_recovery_amount;
double total_recovery_amount;
```



TAX_FORM_LINE

```
/* Usage */
LIST *tax_form_line_list(
        char *tax_form_string,
        int tax_year,
        boolean fetch_account_list,
        boolean fetch_journal_list );

/* Attributes */
char *tax_form_string;
char *tax_form_line_string;
char *tax_form_description;
boolean itemize_accounts;

/* Input */
int tax_year;
```

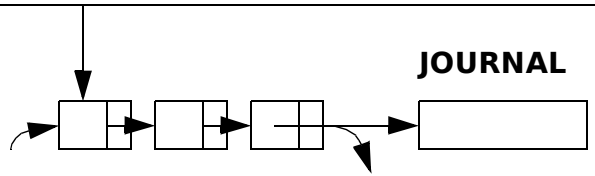
```
/* Process */
LIST *tax_form_line_account_list(
        tax_form_string,
        tax_form_line_string,
        tax_year,
        fetch_journal_list );

/* Operations */
void tax_form_line_list_steady_state(
        tax_form_line_list );

double tax_form_line_total(
        tax_form_line_account_list() );
```



TAX_FORM_LINE_ACCOUNT

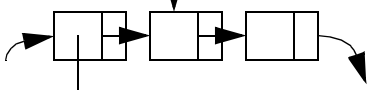
TAX_FORM_LINE_ACCOUNT

```
/* Usage */
LIST *tax_form_line_account_list(
        char *tax_form_string,
        char *tax_form_line,
        int tax_year,
        boolean fetch_journal_list );

/* Attributes */
char *tax_form_string;
char *tax_form_line;
char *account_name;

/* Process */
ACCOUNT *account =
        /* Sets account->accumulate_debit */
        account_fetch(
            account_name );

boolean current_liability =
        subclassification_current_liability(
            account->subclassification_name );

LIST *journal_list =
        journal_year_list(
            tax_year,
            account_name,
            1 /* fetch_transaction_memo */ );

/* Operations */
double tax_form_line_account_total(
        journal_list,
        account->accumulate_debit,
        current_liability );
```



TAX_RENTAL_PROPERTY

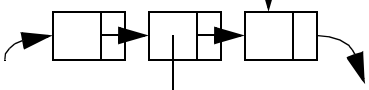
```
/* Usage */
LIST *tax_rental_property_list_fetch(
        char *tax_form_string,
        int tax_year );

/* Attributes */
char  *property_street_address;

/* Process */
LIST *tax_rental_property_string_list(
        tax_rental_property_list );

LIST *tax_form_line_rental_list(
        tax_form_line_list(),
        tax_rental_property_string_list );

double recovery_amount;
double revenue_total [23a];
double mortgage_interest_total [23c];
double depreciation_total [23d];
double expense_total [23e];
double positive_net_income [24];
double loss_total [25];
double tax_rental_property_total();
```



TAX_FORM_LINE_RENTAL

```
char *tax_form_line;
char *tax_form_description;

enum tax_form_line_category;
LIST *tax_form_line_account_list();

LIST *rental_property_list;
double tax_form_line_rental_property_total();
```

revenue
mortgage_interest_paid
depreciation_expense
operating_expense
loss

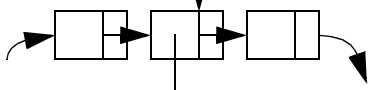


TAX_FORM_LINE_ACCOUNT

INVESTMENT_ACCOUNT

```
ENTITY *financial_entity =
        entity_new(
            full_name,
            street_address );

char *account_number;
char *investment_classification;
char *investment_purpose;

LIST *investment_account_balance_list(
```

```
full_name,
street_address,
account_number );
```



ACCOUNT_BALANCE

```
char *date
double balance;
double balance_change;
int balance_change_percent;
```

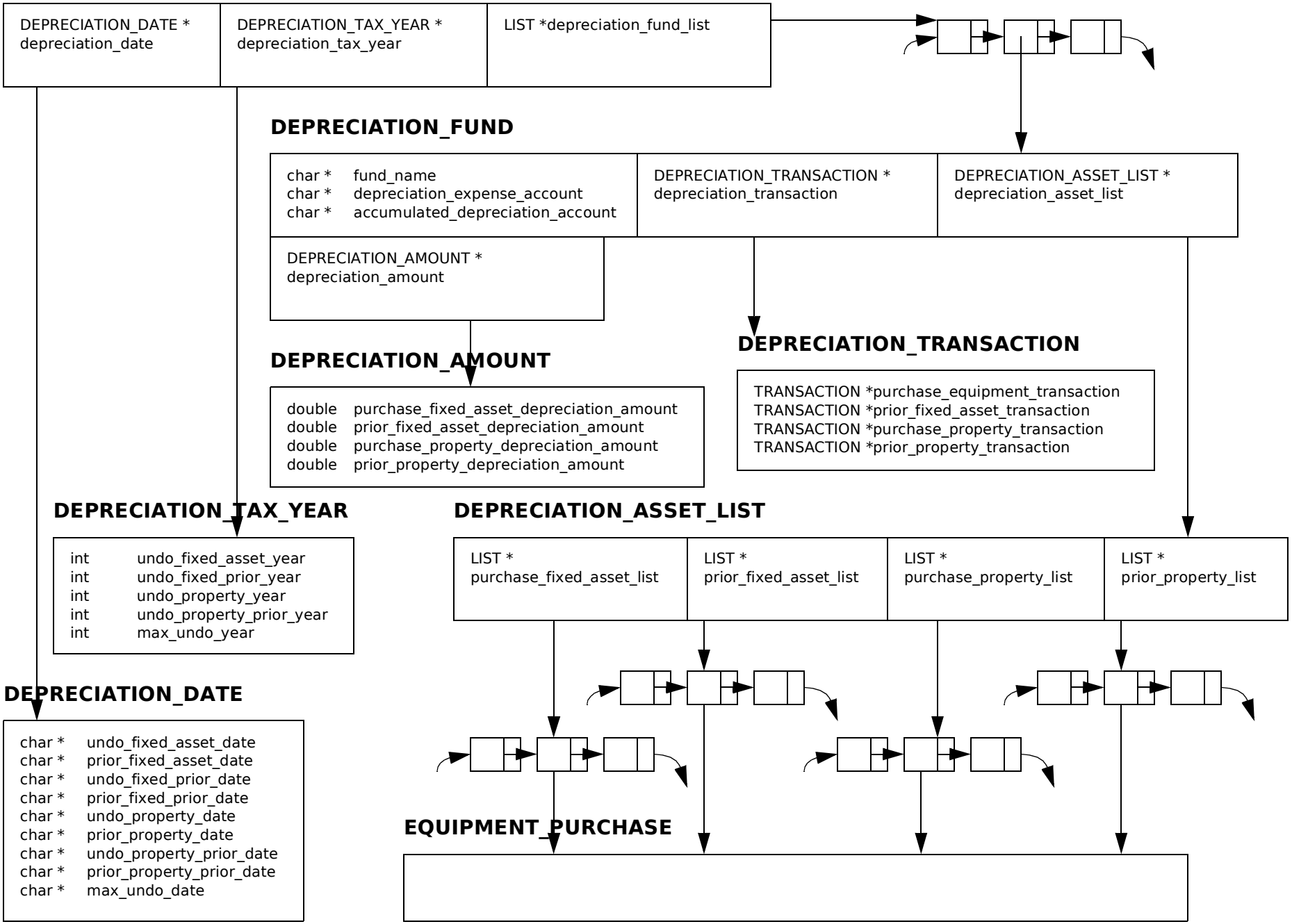
TRANSACTION

ACCOUNT_BALANCE_EQUITY

```
char *date_time
char *investment_operation
double share_price
doubel share_quantity_change
double share_quantity_balance
double cash_in
double market_value
double book_value_change
double book_value_balance
```

```
double moving_share_price
double unrealized_gain_balance
double unrealized_gain_change
double reaqlized_gain
boolean table_first_row
char *transaction_date_time
```

```
TRANSACTION *
transaction
```

PredictiveBooks Data Structure

DEPRECIATION_STRUCTURE